

Source: Local Subject Media

25CY206 - STATISTICAL FOUNDATION FOR COMPUTING AND DATA SCIENCE

Hours Per Week :	L	T	P	SL	C
	3	0	2	3	4

PREREQUISITE KNOWLEDGE: Basic knowledge in Mathematics and Statistics.

COURSE DESCRIPTION:

This course introduces students to probability theory, statistical inference, and computational statistical techniques essential for engineering, Artificial Intelligence, Machine Learning, Data Science, and scientific computing applications. It provides a rigorous foundation in random variables, probability distributions, sampling theory, estimation, hypothesis testing, regression analysis, and multivariate statistical methods. The course emphasizes statistical reasoning under uncertainty and data-driven decision-making. It also integrates computational tools and AI-based platforms for visualization, simulation, and analysis of statistical models. The course enables students to develop analytical thinking, mathematical modeling skills, and computational proficiency required for modern data-centric engineering systems and intelligent computing applications.

COURSE OBJECTIVES:

Understand fundamental probability theory and random variables.

Apply probability distributions in computing and data science problems.

Develop statistical inference and estimation techniques.

Perform hypothesis testing for engineering decision-making.

Analyze relationships using regression models.

Introduce PCA and time series forecasting for AI systems.

Strengthen computational and AI-assisted statistical skills.

MODULE-1

18L+0T+12P+18SL=48Hours

UNIT 1

FUNDAMENTALS OF PROBABILITY

Introduction to probability and its applications in computing, Probability axioms, Conditional probability and independence, Bayes' theorem and its applications. Prior probability and posterior probability; Bayesian inference; Maximum A Posteriori (MAP) estimation.

UNIT 2

RANDOM VARIABLES & PROBABILITY DISTRIBUTIONS

Random variables: Distribution function, discrete random variable, probability mass function, continuous random variable, probability density function, expectation, variance and their properties

Discrete probability distributions: Uniform, Binomial, Poisson. Continuous probability distributions: Uniform, Normal, Exponential, Weibull, Cauchy.

PRACTICES:

Practice No.	Practice	AI Tools / Platforms
1	Explore axioms of probability by formulating real-life event spaces (coin, dice, card experiments) and verifying probability rules such as non-negativity, normalization, and additivity	ChatGPT, Claude AI

Practice No.	Practice	AI Tools / Platforms
2	Visualize conditional probability using Venn diagrams and real-world scenarios such as disease testing or spam filtering systems	GeoGebra AI, Desmos
3	Analyze Bayes' theorem in decision-making problems such as medical diagnosis and email classification and interpret posterior probability updates	Microsoft Copilot, Perplexity AI
4	Compare discrete and continuous random variables by examining PMF vs PDF representations and interpreting their graphical behaviour	Wolfram Alpha, Symbolab
5	Investigate Binomial and Poisson distributions by modeling real-world systems such as defect detection and arrival processes	ChatGPT, Gemini AI
6	Compute expectation, variance, and higher moments for given datasets and interpret their statistical significance in engineering data	Claude AI, ChatGPT
7	Construct probability trees for multi-stage experiments such as machine failure analysis or multi-step decision systems	Perplexity AI, Gemini AI
8	Simulate random variables from different probability distributions and compare theoretical vs empirical results	Wolfram Alpha, ChatGPT
9	Analyze Normal and Exponential distributions using real-world datasets.	ChatGPT, Gemini AI
10	estimate distribution parameters, visualize probability distributions, and interpret the results for Normal and Exponential distributio	ChatGPT, Gemini AI

SKILLS:

- ✓ *Develop functional web and mobile applications using modern AI coding assistants.*
- ✓ *Design effective prompt engineering workflows for architecture and UI generation.*
- ✓ *Integrate frontend components with backend databases, APIs, and cloud services.*
- ✓ *Debug, test, and safely deploy AI-generated application code.*

MODULE-2

27L+0T+18P+27SL=72Hours

UNIT 1

ESTIMATION AND TESTING OF HYPOTHESIS:

Population and sample, sampling distributions, point estimation and properties of estimators, introduction to Maximum Likelihood Estimation (MLE), and confidence intervals for population parameters. Hypothesis testing: Null and alternative hypotheses, level of significance, p-value, critical value, Type I and Type II errors, Single sample and two sample tests for mean's and variance's and chi-square test-Independence attributes. A/B Testing and its applications in data-driven decision-making and AI-enabled systems.

UNIT 2

CORRELATION & REGRESSION ANALYSIS

Correlation analysis: scatter plots, Pearson's correlation coefficient, and interpretation. Simple linear regression model and assumptions, regression coefficients, fitting and interpretation of the regression line, coefficient of determination (R^2), prediction, and residual analysis. Introduction to data transformations (log and square root) and Introduction to logistic regression for binary classification.

UNIT 3

ADVANCED STATISTICAL METHODS

Introduction to Multivariate Analysis (MVA), multivariate data, covariance and correlation matrices, Principal Component Analysis (PCA), dimensionality reduction, interpretation and applications of principal components. Introduction to Time Series Analysis, time series data, components of a time series, trend estimation using the least squares method, moving averages, and simple exponential smoothing for forecasting.

PRACTICES:

Practice No.	Practice	AI Tools / Platforms
1	Analyze sampling distributions and estimate population parameters	ChatGPT, Perplexity AI
2	Explore Maximum Likelihood Estimation by interpreting likelihood functions for statistical models	Wolfram Alpha, Claude AI
3	Construct and interpret confidence intervals for population parameters in engineering datasets	Microsoft Copilot, Gemini AI
4	Perform hypothesis testing (Z-test, t-test, chi-square test) and interpret statistical significance	ChatGPT, Symbolab
5	Compare parametric tests (Z, t, F, chi-square) based on assumptions and applicability	Wolfram Alpha, ChatGPT
6	Visualize simple linear regression models including regression line and residual distribution	GeoGebra AI, Desmos
7	Analyze logistic regression for binary classification problems such as prediction tasks	Perplexity AI, Gemini AI
8	Apply PCA for dimensionality reduction and interpret principal components in feature space	NotebookLM, ChatGPT
9	Investigate time series data patterns including trend and seasonality in real datasets	Microsoft Copilot, Claude AI
10	Validate statistical inference results and compare theoretical vs computed conclusions	Wolfram Alpha, ChatGPT

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to

CO No.	Course Outcomes	Blooms Level	Mapping with POs
1.	Understand the fundamental concepts of probability, random variables, and probability distributions and apply them in computing applications.	Apply	1,2,12
2.	Analyze different probability distributions, their properties, and significance in real-world applications, including computing and machine learning.	Analyze	1, 2, 4
3.	Perform statistical inference using estimation and hypothesis testing techniques, and interpret the results effectively.	Evaluate	2,3,5

CO No.	Course Outcomes	Blooms Level	Mapping with POs
4.	Apply regression analysis techniques, including linear and logistic regression, and interpret results.	Apply	2,4,6
5.	Evaluate the application of Principal Component Analysis (PCA) and Time Series Analysis for data-driven decision-making.	Evaluate	3,5,11

MAPPING OF SUSTAINABLE DEVELOPMENT GOALS (SDGS) & IKS:





TEXT BOOKS:

- Sheldon M. Ross, Introduction to Probability and Statistics for Engineers and Scientists, 5th Edition, Academic Press, 2019.
- Jay L. Devore, Probability and Statistics for Engineering and the Sciences, 9th Edition, Cengage Learning, 2015.

REFERENCE BOOKS:

- Morris H. DeGroot, Mark J. Schervish, Probability and Statistics, 4th Edition, Pearson, 2011.
- Casella, George, Roger L. Berger, Statistical Inference, 2nd Edition, Cengage Learning, 2002.
- Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
- SKILLS
- Statistical reasoning and probabilistic modeling
- Analytical thinking for engineering problem solving
- Mathematical modeling of real-world systems
- Data-driven decision-making ability
- Machine learning and AI-assisted statistical analysis
- Regression and predictive modeling techniques
- Dimensionality reduction and feature engineering
- Time series forecasting and trend analysis
- Python-based scientific computing skills
- Data visualization and interpretation skills
- Image source: https://www.google.com/search?q=PROBABILITY+%26+STATISTICS+normal+graph+IMAGE
- Image file name: PROBABILITY & STATISTICS